Lucas Neumann

Senior Software Engineer — Pricing, Risk & AI Tooling
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Summary

Five years building pricing and risk systems on both sides of the market: a hedge fund, Bank of America's FICC emerging markets desk and an investment manager. Rates, credit and FX derivatives, Bloomberg MARS, vendor PMS platforms, VaR, expected shortfall and Greeks — backed by strong Python, Go and cloud engineering.

Experience

BWGI · Senior Software Engineer 2025–present

- Took **Bloomberg MARS** front-office pricing from proof of concept to daily production, delivering better-quality prices for product classes the legacy PMS did not cover and removing lock-in to that vendor. Automated end to end (portfolio generation, SFTP/S3 delivery, report ingestion) with regression tests and Datadog monitoring.
- Built MARS mappings for **8+ instrument families** — IRS (cleared, OIS, CPI), CDS, TRS, FX forwards/NDFs, barrier and digital options, bonds, CDs and repos — handling quote conventions, face-amount scaling and composite tickers.
- Integrated vendor platforms — Bloomberg (MARS, Data License), Arcesium PMS and other third-party data and pricing products — through complex, high-throughput ETL pipelines, with automated reconciliation against legacy pricing.
- Designed a new risk data pipeline — risk-engine output → PostgreSQL → Qlik — with Greeks and risk-factor serving tables and a data-quality gate (cross-layer reconciliation, robust z-score outliers), run daily per fund.
- Primary on-call for production pricing and risk systems.

Bank of America · Software Engineer, AVP 2023–2025

FICC Emerging Markets. Joined via BRQ as consultant in 2023; hired directly as AVP in 2024.

- Built and ran real-time curve infrastructure for the EM derivatives and FX trading desk: live market data and trader curve marks reaching the sell-side pricing system **within seconds**, plus intraday and EOD curve publication to internal databases; onboarded new engineers to the curve stack.
- Owned the real-time **PTAX** integration with Banco Central do Brasil — a high-availability service with near-zero fault tolerance, where any failure carries direct reputational risk.
- Supported FX and derivatives traders day to day: rates and inflation curve frameworks, bond pricing tools and production troubleshooting.

Apex Capital · Software Developer 2021–2022

Hedge fund

- Worked across the full stack: database administration, data analysis and research reports alongside the analysts, web scraping for data mining, Streamlit risk dashboards and automation of Excel-based operations workflows.
- Migrated the fund's scheduled jobs from Windows Task Scheduler to Apache Airflow on AWS, using modern engineering practices.

Skills

Engineering
Python, Go, SQL (PostgreSQL, Snowflake), MongoDB, Redis, Airflow/Astronomer, Docker, Kubernetes, AWS, Datadog, MCP, Linux, Git

Financial systems
Bloomberg MARS & Data License, Arcesium, MSCI Barra, S&P Global, Qlik

Domain
Rates, credit, FX, fixed income; derivatives pricing; VaR, expected shortfall, Greeks

Languages
Portuguese (native), English (fluent)

Education

UNICAMP — University of Campinas
MSc, Physical Chemistry
2018–2020
Computational simulation of proteins; daily work with HPC clusters, Python, Julia, C, C++ and Linux.

Projects

git-archaeologist (Rust)
Fast CLI extracting time series from git history (LOC, cohort survival, coupling, hotspots) as TSV, CSV, JSON or Parquet.

visidata-rs (Rust)
Rewrite of the VisiData terminal data explorer in Rust (Polars, Ratatui), Lua-extensible like Neovim; in progress.

StringArt.jl (Julia)
Generative art: turns photos into string-art patterns.

MonteCarloArt.jl (Julia)
Generative art: pointillist images via Monte Carlo sampling.